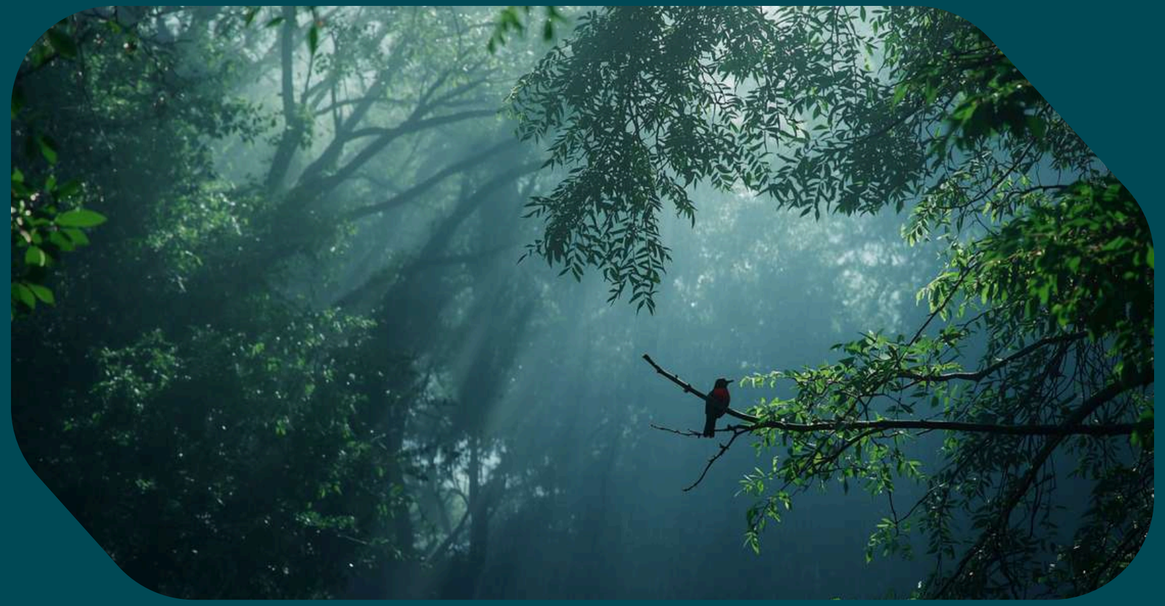

# Wildlife Species Classification Using Audio Signals

## BirdCLEF Dataset Project

Leveraging deep learning and the BirdCLEF dataset to identify bird species through audio analysis for biodiversity monitoring.

Presented by:
Attem Deepak Mudiraj

BirdCLEF Project

Introduction

# Importance of Biodiversity Monitoring

Understanding and protecting wildlife species is essential for maintaining healthy ecosystems worldwide.

- Biodiversity is crucial for ecosystem health and stability
- Traditional wildlife observation methods are challenging and time-consuming
- Audio signals enable non-invasive species monitoring
- Automated classification supports conservation efforts

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## Problem Statement

**CHALLENGES:**

Manual monitoring is labor-intensive and time-consuming
Human identification is prone to errors
Limited scalability for large ecosystems

**SOLUTIONS:**

Automated classification using deep learning
Scalable processing of audio recordings
Consistent and accurate species identification

# Challenges & Solutions

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# Project Objectives

## Goals & Targets

Three core objectives driving our wildlife audio classification project.

Convert raw audio to spectrogram images
Train deep learning models for species ID
Achieve high classification accuracy
Validate on BirdCLEF dataset

# BirdCLEF Dataset

Wildlife Audio

## Dataset Overview

Sourced from Kaggle BirdCLEF challenge with thousands of labeled audio recordings.

Focus on top 20 bird species classes
Thousands of high-quality audio samples
Diverse recording environments
Labeled by expert ornithologists

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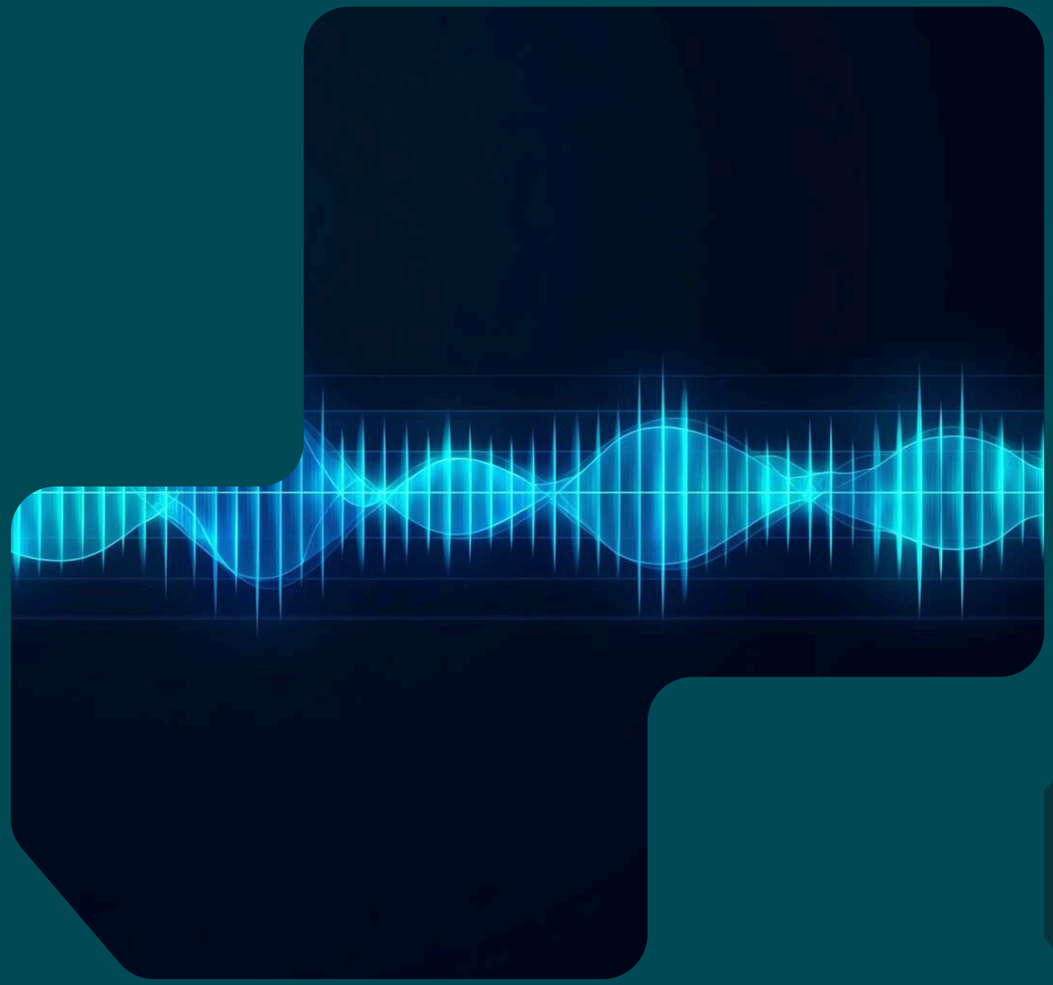

# Data Preprocessing

## Preparing Audio for Analysis

Raw audio files are transformed into clean, standardized segments for model training.

- Audio loading using Librosa library
- Segmenting audio into 5-second clips
- Noise reduction techniques applied
- Silence trimming for cleaner data

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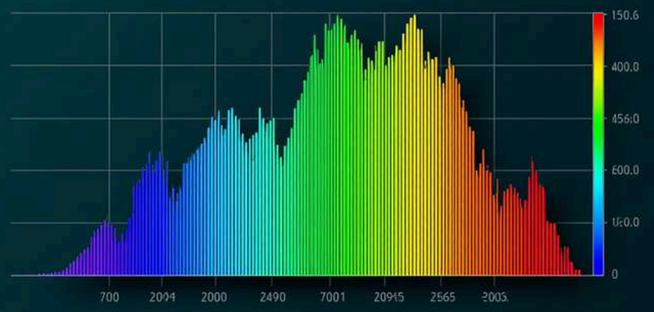
## Understanding Audio Features

**SPECTROGRAM:**
Converts audio into time-frequency images
Visualizes sound patterns over time
Enables CNN-based image classification

**MEL SCALE:**
Aligns with human hearing perception
Better captures relevant audio features
Improves model accuracy for bird calls




# Feature Extraction: Mel Spectrogram

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# Model Architecture

EfficientNet
CNN

Leveraging pretrained convolutional neural networks for bird species classification.

Pretrained CNN backbone
Transfer learning with fine-tuning

High accuracy on spectrograms
Computationally efficient architecture

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# Proposed Approach

## Multi-Segment Learning Strategy

Robust classification through segment aggregation and balanced training.

Use 3 segments per audio for robust learning
Aggregate segment predictions for final output

Apply class weighting to handle imbalance
Improves minority species detection accuracy

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# Training Setup

## Model Training Parameters

### Details:

Loss Function: CrossEntropyLoss for multi-class classification
Optimizer: Adam with adaptive learning rate
Epochs: 5 training cycles
Batch Size: 32 samples per iteration

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## Classification Results

**ACCURACY:**

Accuracy: 70-85% range achieved
Precision scores across species classes
Recall metrics for minority classes

**VALIDATION:**

Strong performance on common species
Improved minority class detection
Consistent validation results

# Model Performance

# Data and Model Visualizations

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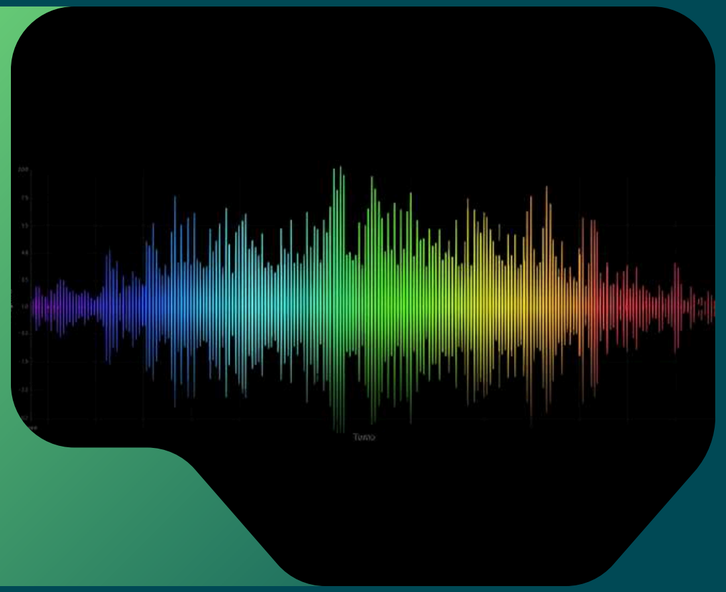
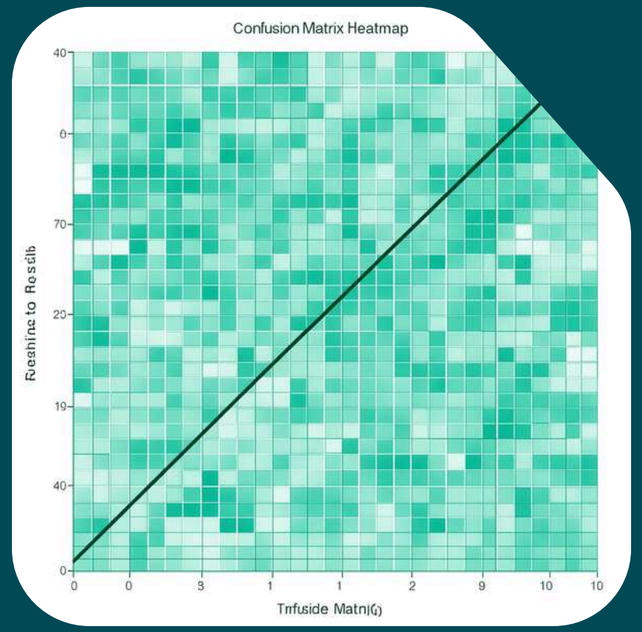


## Key Visual Outputs from the Project

Visual representations help interpret audio data and model performance effectively.




Audio waveform visualization
Mel spectrogram representation
Training loss curve over epochs
Confusion matrix heatmap

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# Challenges

## Obstacles Encountered

Key challenges faced during the wildlife audio classification project.

- Background noise in recordings
- Class imbalance in dataset
- Multiple bird species in single clips
- Variation in recording quality

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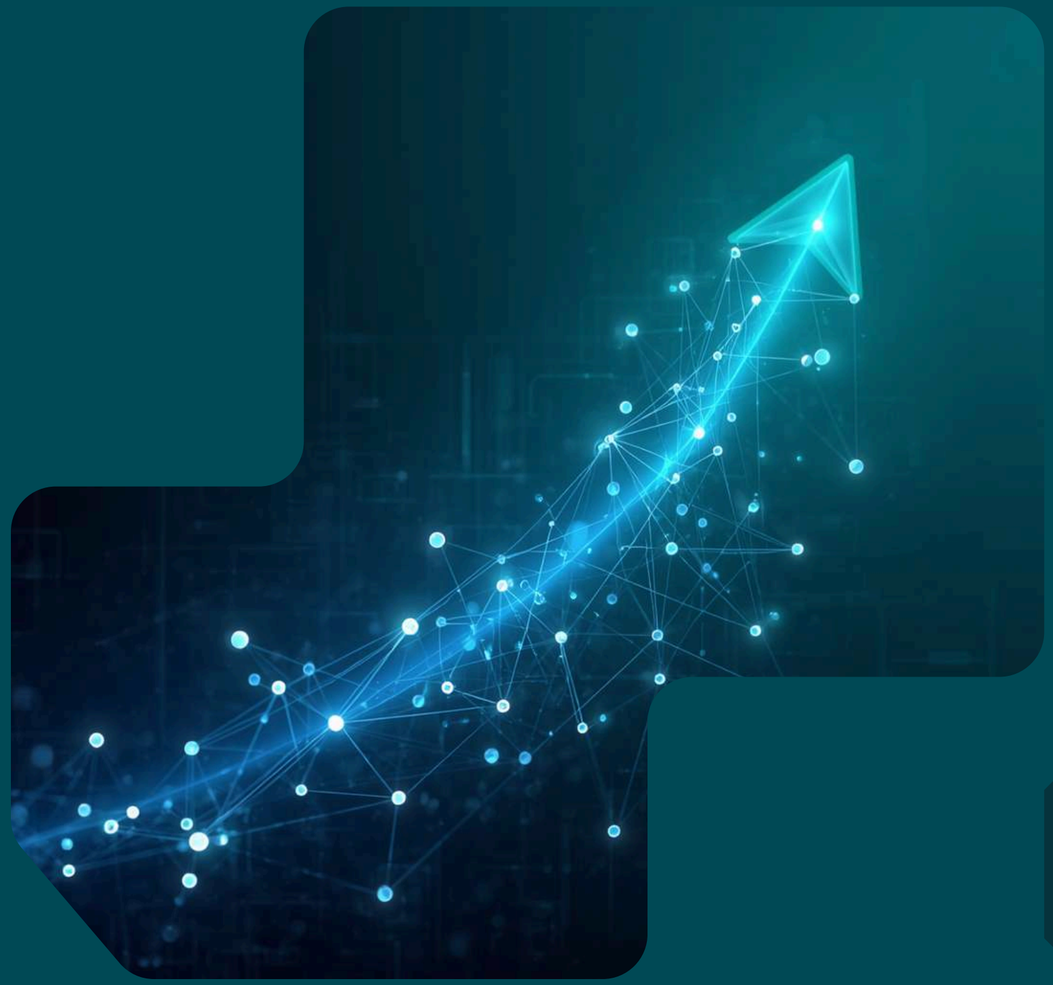

# Key Improvements

## Enhancements Made

Strategic improvements to boost model accuracy and robustness.

Pretrained EfficientNet for better feature extraction
Multi-segment approach for data robustness

Class balancing techniques applied
Improved minority class detection rates

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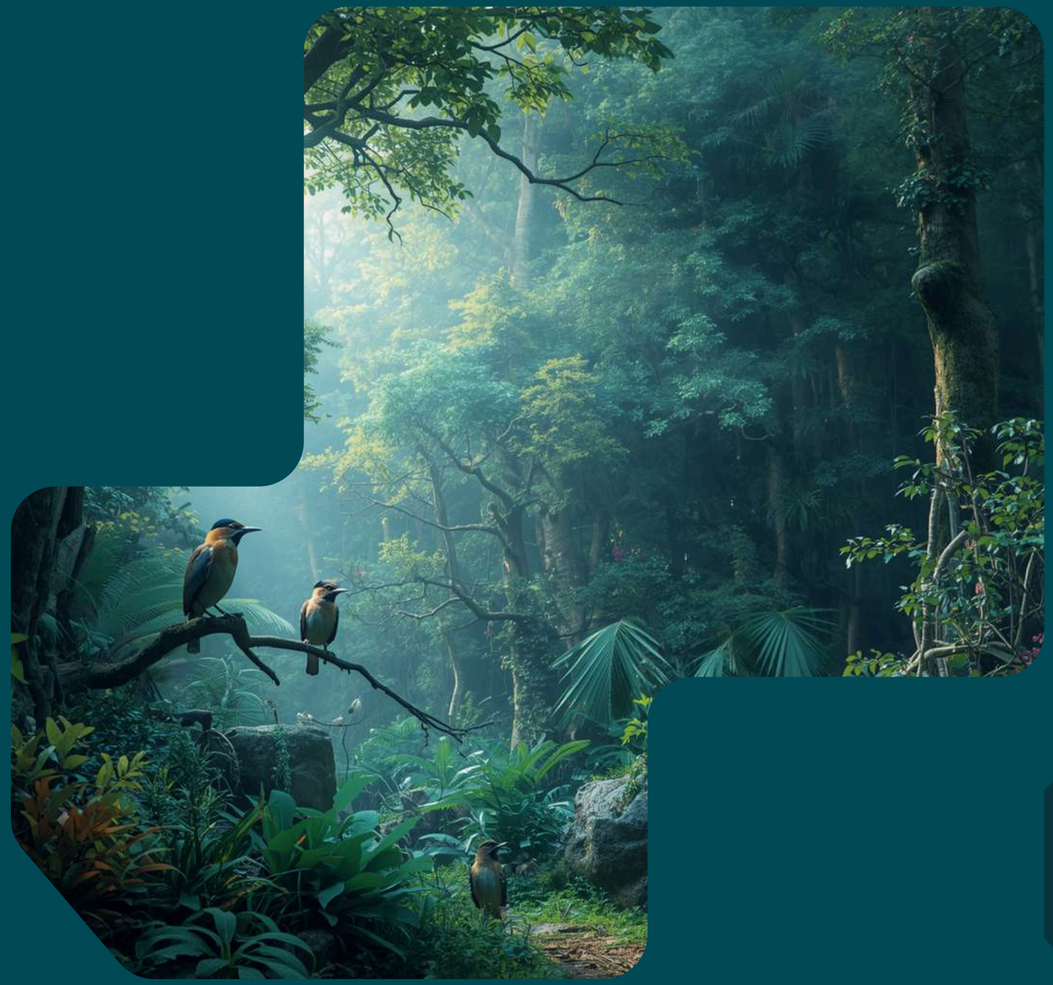

# Applications

## Real-World Impact

Practical applications of audio-based wildlife classification technology.

Wildlife conservation monitoring
Ecosystem health assessment

Smart sensor biodiversity tracking
Automated species detection networks

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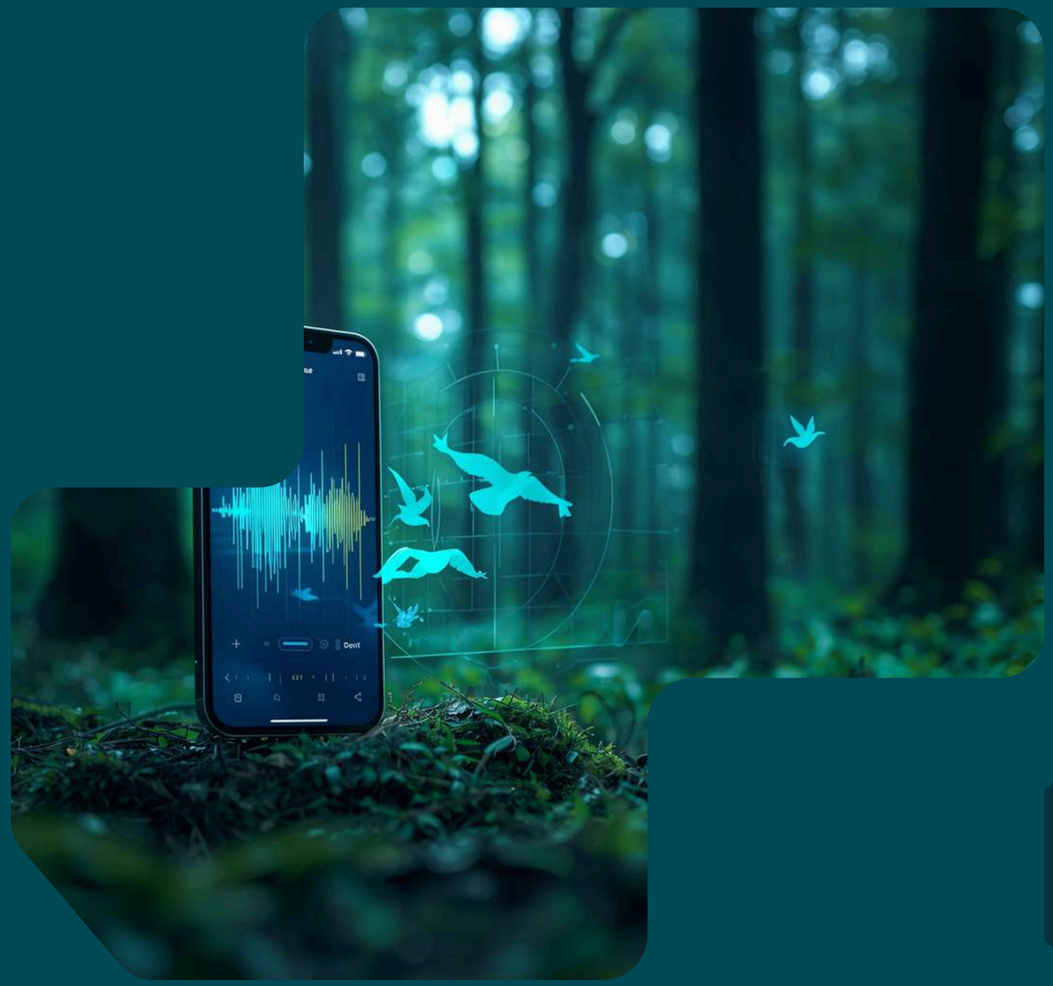

# Future Directions

## Expanding the Horizon

Potential project expansions to enhance wildlife audio classification systems.

Multi-label classification for overlapping species
Real-time audio detection systems

Development of mobile app for field use
Edge computing for remote deployment

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# Conclusion

## Project Summary

Deep learning enables accurate and scalable wildlife audio classification.

Successful bird species classification using audio
Demonstrated deep learning effectiveness on BirdCLEF

Foundation for scalable biodiversity monitoring
Non-invasive wildlife conservation approach

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> Your engagement with wildlife audio classification makes a difference.

# Thank You!

## References & Resources


Kaggle BirdCLEF Dataset
Librosa Audio Library
PyTorch Framework